

Original article

Seroprevalence of *Toxoplasma gondii* and associated factors among pregnant women consulted in Buhodle Healthcare Centers, SomaliaMohamed Osman^a, Abdullahi Adan^b, Abdulahi Abdiwali Mahamed^{a,c}, Md Al-Mamun^{d,e,*}, Marufa Alam^e^a Geedi Somali Research Institute, Somalia^b Jigjiga University, College of Health and Medical Science, Department of Veterinary Microbiology and Public Health, Jigjiga, Ethiopia^c Kayd Consultancy, Somalia^d BRAC Institute of Governance and Development (BIGD), BRAC University, Dhaka, 1212, Bangladesh^e Department of Public and Community Health, Faculty of Medicine and Health Sciences, Frontier University, Garowe, Puntland, Somalia

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ABSTRACT

Problem considered: *Toxoplasma gondii* infection is a globally prevalent parasitic disease with serious health consequences, especially for pregnant women due to the risk of congenital transmission. Despite its importance, limited data exist on its prevalence in Somalia. This study aimed to determine the seroprevalence of *T. gondii* infection and identify associated risk factors among pregnant women attending healthcare centers in Buhodle District, Somalia.

Methods: A cross-sectional study was conducted from March to May 2024 among 384 randomly selected pregnant women. Serum samples were analyzed using competitive ELISA (cELISA) to detect *T. gondii* antibodies. Data were processed using EpiData and SPSS. Descriptive statistics summarized participant characteristics, while bivariate and multivariate logistic regression analyses identified associations between potential risk factors and seropositivity. A p-value <0.05 and a 95 % confidence interval were considered statistically significant.

Results: The overall seroprevalence of *T. gondii* was 36.4 % (140/384). Among these, 28.1 % (108/140) were positive for anti-*T. gondii* IgG antibodies, indicating prior exposure, while 8.3 % (32/140) were positive for IgM, reflecting recent infection. Multivariable logistic regression revealed significant associations between *T. gondii* seropositivity and lower educational attainment, as well as frequent contact with soil ($p < 0.05$).

Conclusion: The high prevalence of *T. gondii* infection among pregnant women in Buhodle District indicates a substantial public health concern. Targeted community awareness programs, enhanced antenatal screening, and further epidemiological and molecular studies are essential to reduce maternal and fetal health risks.

1. Introduction

Toxoplasmosis is a disease caused by an obligate intracellular protozoan parasite *Toxoplasma gondii* (*T. gondii*). Numerous bird and mammalian species have been found to be infected, and the illness has been documented in the majority of the world's regions.¹ The organism is regarded as an incredibly effective parasite, since it rarely harms its host.² Toxoplasmosis was first discovered as an intracellular parasite of the spleen and other organs of the African rodent (*Chenodactylus gundi*), a growing amount of attention has been focused on the disease because it has been identified as a pathogen in numerous hosts, including

humans.³ *Toxoplasma* is a parasite that seems to be universally distributed and non-specific in warm-blooded animals. The central nervous system and reticulo-endothelial tissues are its preferred sites. On multiplication, a mass of the parasites within a single host cell can arise from multiple divisions caused by longitudinal fission.⁴ Then a strong protective membrane known as a pseudocyst form around the parasite aggregation¹ (see Fig. 1).

Since cats are the only known definitive hosts for *Toxoplasma gondii*, they serve as the primary infection reservoirs. *Toxoplasma gondii* infects host cells through three stages: sporozoites, bradyzoites, and tachyzoites.³ Toxoplasmosis can be contracted congenitally, through eating

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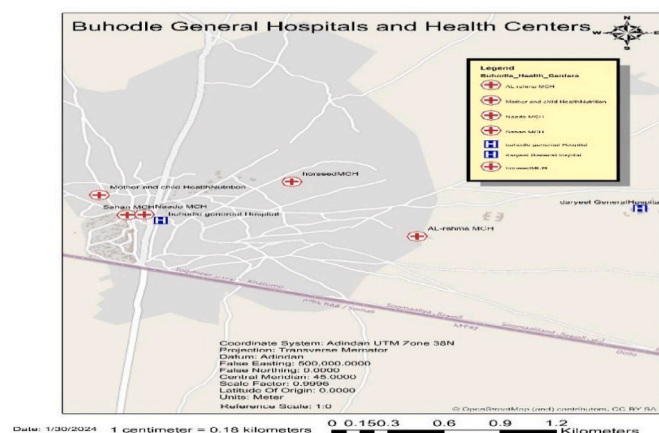


Fig. 1. Map of the study area, Buhodle, Somalia.

feces containing oocysts, or through carnivorousness.² Both domestic and wild cats are susceptible to both systemic and patent intestinal infections; however, the parasite only causes systemic infections, which usually lead to tissue cyst formation, in intermediate hosts such as humans and other animals.³

Human infections that happen during pregnancy are especially dangerous because they can lead to congenital disorders, which mainly affect the central nervous system, or abortion. Workers in abattoirs and other individuals who handle raw meat are frequently exposed to the parasite.^{3,5} Due to cultural differences regarding exposure to undercooked meat, soil, and cats, the significance of these different risk factors probably varies significantly between residential areas.⁴

Although it rarely results in abortion in cattle or horses, *Toxoplasma gondii* is a significant abortifacient in sheep and goats.⁵ The Centers for Disease Control and Prevention⁶ has identified *T. gondii* infection as a neglected parasitic disease that results in fetal loss and congenital abnormalities in both humans and animal.⁷ There are differences in the severity and rate of transmission of congenital *T. gondii* infection based on trimester.⁸ On the other hand, the third trimester has the highest transmission rate and the lowest infection severity.⁹

Approximately one-third of people worldwide are at risk of contracting the infection due to parasite exposure. On the other hand, reports have indicated that 400–4000 cases of congenital toxoplasmosis occur annually.¹⁰ Numerous countries around the world have reported on its prevalence: in the United States, 15 % of women of childbearing age, in France (44 %), in Indonesia (above 60 %), in Latin America (30–75 %), in parts of Eastern/Central Europe, the Middle East, South East Asia, and Africa, among pregnant women.^{2,8,11} Over the previous three to four decades, the prevalence has decreased in some areas of Eastern and Central Europe.¹² Women in parts of south-east Asia, the Middle East, and Latin America who are pregnant or have children have been reported having a high prevalence of the infection.¹¹

In Africa, especially in Sub-Saharan Africa, there is a higher prevalence of *T. gondii* in relation to HIV.¹³ Various African nations have reported varying prevalence rates, including Sudan at 79.8 %¹⁴ and Ghana at 92.5 %.¹⁵ Few pregnant women contract the infection; the majority of *T. gondii* infections are chronic. On the other hand, there is a chance that a pregnant woman who contracts an acute infection during her pregnancy will congenitally infect the fetus.¹⁶ In Somalia, the prevalence ranges from 45.2 %¹⁷ to 51.79 %¹⁸ from the general population with no data among HIV patients. The overall data studied about the infection in Somalia is very narrowed to these two studies. As far as we are aware, serological screening for *T. gondii* infection is not a routine procedure for pregnant women. Furthermore, the studies areas lack documented data regarding the disease's prevalence and related risk factors.^{9,16,18} Antenatal data on this parasitic disease are thought to be useful for planning and implementing comprehensive *T. gondii* diagnostic, control, and

prevention strategies, as well as providing baseline information about the prevalence of *T. gondii* in pregnant women in the study area. Therefore, this is aimed to determine seroprevalence of *T. gondii* and associated factors among pregnant women who consulted healthcare center in Buhodle district, Somalia.

2. Materials and methods

2.1. Study area

The study was conducted in Buhodle district which is located in around 900 km north of Somalia's capital, Mogadishu, and with a total surface of 9.98 km² (1, 236 ha). The district has an estimated total population of 201, 747.¹⁹ According to Somalia Demographic and Health Information Survey 2020, there are more than five public healthcare facilities, one district general hospital and four health centers, in the district and more than fifteen other healthcare facilities that comes under the district's administration in the surrounding areas. Majority of the area residents have an access to healthcare facilities, both public and private health sectors.¹⁷

The surrounding residents and sub-districts are rich in livestock with growing agricultural activities. Somalia has an estimated population of 43 million livestock. The livestock sector is the largest contributor to Somali livelihoods with over 65 % of the population engaged in some way in the industry.²⁰ Buhodle is located at an elevation of none meters (0 feet) above sea level, Buhodle has a Subtropical desert climate. The district's yearly temperature is 28.22 °C (82.8 °F) and it is 0.31 % higher than Somalia's averages.

2.2. Study design

A cross-sectional study was conducted from March 2024 to May 2024 to determine the prevalence and to identify the associated risk factors of toxoplasmosis among pregnant women consulted to healthcare facilities.

2.3. Source population

All pregnant women living in Buhodle district during the study period are the source population.

2.4. Study population

The study population was all pregnant women who live in Buhodle who consulted healthcare facilities.

2.5. Inclusion criteria

Those pregnant women who attend antenatal consultations at healthcare facilities in Buhodle district.

2.6. Exclusion criteria

Those pregnant women who are unable or refused to participate the study during data collection, as well as those patients who were on antibiotic treatment.

2.7. Sample size determination

The required sample size was calculated using the formula from Thrustfield, 2007 with prevalence of as 50 % and 95 % CI and 5 % margin of error. So, the minimum sample size was calculated with the formula:

$$n = z^2 x p q / d^2$$

Where, n = sample size?

z = confidence level which is 95 % - 1.96

P = 50 % - 0.5

q = $1-p$ = 0.4821

d = margin of error 5 % = 0.05

$$n = \frac{(1.96)^2 (0.5) (0.4821)}{(0.05)^2}$$

n = 384

2.8. Sampling techniques

Both the district and the hospital were purposively selected due its accessibility and availability of required diagnostic tests throughout the region. During the study, list of pregnant women attending were obtained from the desk information of the hospital, from this sampling frame were obtained and the sampled pregnant women was selected using random sampling technique.

2.9. Ethical consideration

The study protocol was reviewed and approved by the research ethics and reviewer committee of jigjiga university collage of health and medical science with Ref No IHRRCO3/2024. Official permission was also obtained from the respective body at Buhodle General hospital and other health care facilities by providing letters from the university. Verbal written informed consent was obtained from each participant after informing about the purpose of the study.

2.10. Blood sample collection

Blood samples were collected from study participants who were visiting those healthcare facilities during the study period. After written consent was gained from the participants; 3–5 ml of venous blood was collected with plain tube by strictly following standard operational procedure.

2.11. Handling and sample transport

The serum was separated from the other blood compartments after clotting by centrifugation at 5000 rpm for 10 min. Samples collected from health care facilities other than the general hospital was transported with ice cold chain to the site of investigation. The serum samples were kept at -20°C till serological test are done.

2.12. Sociodemographic data

Information for sociodemographic data, history of exposure for the possible associated factors, and other valuable information was collected using structured questionnaire.

2.13. Laboratory investigation

The *Toxoplasma gondii* antigens were applied to a pre-coated microtiter plate. As directed by the manufacturer, human serum samples were diluted 1:100 along with positive and negative controls. First, a blocking buffer was used to block nonspecific binding sites. Next, diluted serum samples were added to the wells. An enzyme-conjugated secondary antibody (anti-human IgG or IgM) was added after incubation, and the plate was then incubated once more. After adding the enzyme's substrate, a stop solution was used to halt the reaction. A microplate reader was used to measure each well's optical density (OD) at 450 nm.

The negative control had low OD values, whereas the positive control had high OD values, suggesting the presence of antibodies. Serum samples were deemed positive for *Toxoplasma gondii* antibodies if their OD values exceeded the cutoff value, which was calculated by adding

the negative control by two standard deviations. Samples with OD values below the cutoff were deemed negative. The assay's accuracy was confirmed by the results, which were in line with what was anticipated for the controls.

2.13.1. Test principles and protocols

2.13.1.1. Test principle. The Atlas *Toxoplasma* IgG and IgM ELISA Test Kit are a solid phase enzyme immunoassay based on indirect principle for the qualitative detection of IgG antibodies to *Toxoplasma* in human serum or plasma. The microwell plate is coated with *Toxoplasma* recombinant antigens. During testing, the specimen diluent and the specimens are added to the antigen coated microwell plate and then incubated. If the specimens contain IgG or IgM antibodies to *Toxoplasma*, it will bind to the antigens coated on the microwell plate to form immobilized antigen-*Toxoplasma* IgG and IgM antibody complexes. If the specimens do not contain IgG or IgM antibodies to *Toxoplasma*, the complexes will not be formed. After initial incubation, the microwell plate is washed to remove unbound materials. The enzyme-conjugated anti-human IgG and IgM antibodies are added to the microwell plate and then incubated. The enzyme-conjugated anti-human IgG or IgM antibodies will bind to the immobilized antigen-*Toxoplasma* IgG or IgM antibody complexes present. After the second incubation, the microwell plate is washed to remove unbound materials. Substrate A and substrate B are added and then incubated to produce a blue color indicating the amount of *Toxoplasma* IgG or IgM antibodies present in the specimens. Sulfuric acid solution is added to the microwell plate to stop the reaction producing a color change from blue to yellow. The color intensity, which corresponds to the amount of *Toxoplasma* IgG and IgM antibodies present in the specimens, is measured with a microplate reader at 450/630–700 nm or 450 nm.

2.13.1.1.1. Materials required.

- *Toxoplasma* IgG Microwell Plate
- *Toxoplasma* IgG and IgM Conjugate
- Concentrated Wash Buffer (25x)
- Specimen Diluent
- Substrate A
- Substrate B
- Stop Solution
- *Toxoplasma* IgG and IgM Negative Controls
- *Toxoplasma* IgG and IgM Cut-off
- *Toxoplasma* IgG and IgM Positive Controls
- Plate Sealers
- Distilled water
- Decontamination solution
- Paper towel
- Microwell plate washer

- Gloves
- Micropipettes and tips
- Graduated cylinders
- Mixer
- Timer
- Microplate reader

2.13.1.1.2. Test procedure. Allow reagents and specimens to reach room temperature (15–3°C) prior to testing. The procedure must be strictly followed. Assay must proceed to completion within time limits. Arrange the controls so that well A1 is the Blank well. From well A1, arrange the controls in a horizontal or vertical configuration. The procedure below assigns specific wells arranged in a vertical configuration. Configuration may depend upon software.

2.14. Data quality assurance

A clear explanation of the purpose and procedures of the study was given to introduce each interview to get the right information from each participant. A standard operational procedure (SOP) was strictly followed for sample collection, transport and storage. To avoid the measurement bias that encountered during testing of the serum sample, Internal Quality control was done. Special emphasis was given during coding the data sheet as well as the sample. All the data obtained were checked for the completeness of the information at each time. All reagents that are used for testing were checked for their shelf life, being at appropriate temperature before using them. Test procedures were done according to the manufacturer's instruction.

2.15. Data quality control

To ensure the quality of data, a one-day training session for data collectors and supervisors, a pre-test of the data extraction checklist, and ongoing supervision throughout the data collection were used. The data extraction checklist was pretested on 5 % of the sample outside the selected health facility. The pretesting of the data extraction checklist was done one week before the actual data collection. During the actual data collection, the supervisor reviewed the collected data each day to ensure its completeness and consistency. The data were re-checked during data entry, and any incomplete questionnaires were eliminated.

2.16. Data entry, management and analysis

Data were entered Epi-data version 4.6 and data analysis was done in SPSS version 25. The data was summarized as number and percentage for categorical variables. Bivariate and multi-variable logistic regression was carried out to assess the association between dependent and independent variables. Variables with p-value ≤ 0.25 was considered to be candidates for multi-variable logistic regression. Variables with P-value ≤ 0.05 and 95 % Confidence interval in multi-variable regression was considered statistically significant.

3. Results

3.1. Seroprevalence of toxoplasma gondii among pregnant women

The overall seroprevalence of *T. gondii* among the pregnant women was 140/384 (36.4 %). Further, the prevalence of IgG and IgM Anti-*T. gondii* antibodies were 108/140 (77 %) and 32/140 (23 %) respectively as mentioned in Table 1. The majority 221 (57.6 %) of them between 26 and 40 years of age. Most 300 (78.1 %) of the pregnant women

Table 1

Step-by-step test procedure for the enzyme-linked immunosorbent assay (ELISA).

Step	Detailed Procedure	Simplified Procedure
1	• Prepare Working Wash Buffer by diluting the Concentrated Wash Buffer 1:25. Pour the contents of the bottle containing the concentrated wash buffer in a graduated cylinder and fill it with freshly distilled or deionized water to 1250 mL for 96 wells/plate testing, or 625 mL for 48 wells/plate testing. The Working Wash Buffer is stable for 2 weeks at 15–30°C Note: If crystals are present in the Concentrated Wash Buffer, warm it up at 37 °C until all crystals dissolves. • Remove unused strips from the microwell plate, and store in the original resealable pouch at 2–8 °C.	• Prepare Working Wash Buffer by diluting the Concentrated Wash Buffer 1:25 • Remove and store unused strips at 2–8 °C
2	• Leave A1 as Blank well.	• Leave A1 as Blank well.
3	• Add 100 µL of Negative Control in wells B1 and C1. (bale yellow Reagent) • Add 100 µL of Cut-off 1 in wells D1 and E1. (Blue Reagent) • Add 100 µL of Positive Control in wells F1 and G1. (Blue Reagent)	• B1 and C1: Add 100 µL Negative Control • D1 and E1: Add 100 µL Cut-off • F1 and G1: Add 100 µL Positive Control
4	• Add 100 µL of Specimen Diluent to assigned wells starting at B2. The color of Specimen Diluent is green. • Add 5 µL of specimen to assigned wells starting at B2. Then a color change from green to blue will occur to verify that the specimen has been added.	• Starting B2: Add 100 µL Specimen Diluent • Starting B2: Add 5 µL specimen
5	• Mix gently by swirling the microwell plate on a flat bench for 30 s. • Cover the microwell plate with the Plate Sealer and incubate in a water bath or an incubator at 37C $\pm$ 2C for 30 min $\pm$ 2 min.	• Mix gently • Cover the microwell plate with the Plate Sealer and incubate at 37 °C for 30 min
6	• Remove the Plate Sealer. • Wash each well 5 times with 350 µL of Working Wash Buffer per well, then remove the liquid. • Turn the microwell plate upside down on absorbent tissue for a few seconds. Ensure that all wells have been completely washed and dried. Note: Improper washing may cause false positive results.	• Remove the Plate Sealer • Wash each well 5 times with 350 µL of Working Wash Buffer • Turn the microwell plate upside down on absorbent tissue
7	• Add 100 µL of Conjugate to each well except for the Blank well. The color of Conjugate is red.	• Add 100 µL of Conjugate to each well except for the Blank well
8	• Cover the microwell plate with the Plate Sealer and incubate in a water bath or an incubator at 37C $\pm$ 2C for 30 min $\pm$ 2 min.	• Cover the microwell plate with the Plate Sealer and incubate at 37 °C for 30 min
9	• Repeat Step 4.	• Repeat Step 4
10	• Add 50 µL of Substrate A to each well. (Clear Reagent) • Add 50 µL of Substrate B to each well. (Clear Reagent) Then a blue color should develop in wells containing Positive specimens.	• Add 50 µL of Substrate A to each well • Add 50 µL of Substrate B to each well
11	• Mix gently then cover microwell plate with Plate Sealer and incubate in a water bath or incubator at 37C $\pm$ 2C for 10 min $\pm$ 1 min.	• Mix then cover microwell plate with Plate Sealer and incubate at 37 °C for 10 min
12	• Remove the Plate Sealer.	• Remove the Plate Sealer

(continued on next page)

Table 1 (continued)

Step	Detailed Procedure	Simplified Procedure
13	Add 50 µL of Stop Solution to each well. (Clear Reagent) Then a yellow color should develop in wells containing Positive specimens.	Add 50 µL of Stop Solution to each well
	- Read at 450/630–700 nm in 30 min. - Note: Microwell plate can also be read at 450 nm, but it is strongly recommended to read it at 450/630–700 nm for better results.	Read at 450/630–700 nm in 30 min

were urban residents. Population in the age group of 26–40 had more seropositive 85/221 (38.4 %) when compared to other age groups. In addition, 265 (69 %) of the study participants were multigravida and they had more seropositivity than those women who were on primigravida status 42 (35 %) (see Table 2).

The majority 221 (57.6 %) of them between 26 and 40 years of age. The mean age of the study participants was 28.7 (SD ± 6.853) and the minimum and maximum age were 15 and 43 years respectively. Most 300 (78.1 %) of the pregnant women were urban residents.

As indicated in table below, none of the study participants were HIV patients according to their responses as we didn't perform HIV test for them since it was not part of plan to perform the test. As well as there was no habit of eating raw meat among the pregnant women or it was zero in percent.

Based on trimester status, pregnant women who were on the third trimester 134 (34.9 %) were more than those participants who were on the second and first trimester which were 133 (34.6 %) and 117 (30.5 %) respectively. In addition to that, most of the pregnant women who were on third trimester 51 (38 %) were more seropositive when compared to the first and second trimesters (40 (34 %), 49 (37 %) respectively) (see Table 3).

According to the study participants, those who were illiterate 57/149 (38 %) were more seropositive than those who had primary-secondary

Table 2

Seroprevalence of toxoplasma gondii among pregnant women at Buhodle healthcare center, Somalia.

Variable	Category	Frequency (%)	ELISA result Positive (%)
Age	15–25	152 (39.6)	52 (34.2)
	26–40	221 (57.6)	85 (38.4)
	>40	11 (2.9)	3 (27)
Education	Illiterate	149 (38.8)	57 (38)
	Primary-Secondary	177 (46.1)	48 (27)
	Tertiary education	58 (15.1)	35 (60.3)
	Urban	300 (78.1)	97 (32)
Residence	Rural	84 (21.9)	11 (13)
Gestational age	First trimester	117 (30.5)	40 (34)
	Second trimester	133 (34.6)	49 (37)
	Third trimester	134 (34.9)	51 (38)
Gravidity	Primigravida	119 (31)	42 (35)
	Multigravida	265 (69)	98 (37)
HIV status	Positive	0 (0)	0 (0)
	Negative	384 (100)	140
Eating raw meat	Yes	0 (0)	0 (0)
	No	384 (100)	140 (36)
Abortion history	Yes	49 (13)	23 (47)
	No	335 (87)	117 (35)
Cats at home	Yes	157 (41)	67 (43)
	No	227 (59)	73 (32)
Soil contact	Yes	202 (53)	120 (59)
	No	182 (47)	49 (26)
Transfusion history	Yes	78 (20)	28 (36)
	No	306 (80)	112 (37)

Anti-T. gondii antibodies among pregnant women at Buhodle healthcare center.

Table 3

Bivariate and logistic regression analysis from toxoplasmosis of pregnant women at Buhodle healthcare Center, Somalia.

Variable	Category	ELISA result Positive (%)	COR (95 %CI)	P-Value
Age	15–25	52 (34.2)	1.387 (0.4–5)	0.640
	26–40	85 (38.4)	1.667(0.4–6)	0.460
	>40	3 (27)		
Education	Illiterate	57 (38)	4.090(2–7)	0.000
	Primary-Secondary	48 (27)	1.665(1–2)	0.033
	Tertiary education	35 (60.3)		
Occupation	Employed	43 (34)	0.860(0.5–1)	0.507
	Unemployed	97 (37.5)		
Residence	Urban	113 (38)	0.784(0.5–1)	0.353
	Rural	27 (32)		
Trimester	First	40 (34)	0.845(0.5–1)	0.525
	Second	49 (37)	0.949(0.6–1.6)	0.837
	Third	51 (38)		
Gravida	Primigravida	42 (35)	1.076(0.7–1.7)	0.751
	Multigravida	98 (37)		
Abortion history	Yes	23 (47)	0.607(0.3–1)	0.105
	No	117 (35)		
Cats at home	Yes	67 (43)	1.637(1.3–2.1)	0.036
	No	73 32(.)		
Soil contact	Yes	87 (43)	1.45 (1.2–37)	0.014
	No	53 (41)		
Transfusion history	Yes	28 (36)	1.031(0.6–1)	0.908
	No	112 (37)		

CI = Confidence Interval COR = Crude Odd Ratio.

and tertiary education.

3.2. Bivariate logistic analysis for potential associated risk factors

Bivariate and multivariate logistic regression was used to assess if there was any association between the independent variables and dependent variable. In bivariate logistic regression analysis in identified the following risk factors education, abortion history, presence of cats at home, soil exposure, became candidates for multivariable logistic regression ($p \leq 0.25$). However, only illiterate and soil contact have shown significant association in multivariate regression.

3.3. Multivariable logistic analysis for associated risk factors

Variables which became candidates for multivariable logistic regression were taken and analyzed. Therefore, in multivariable logistic regression analysis, illiteracy (AOR: 4.036 (95 %CI 2–7.6), $p = 0.000$), and soil contact (AOR: 1.45, 95 %CI (1.2–3.7), $p = 0.014$), were found to be statistically significant ($p \leq 0.05$ with 95 %CI). Pregnant women who had no education were 4.036 (AOR:4.036 95 % CI:2.124–7.666) times more likely to develop toxoplasmosis than those who had tertiary education. Regarding soil exposure, those participants who had contact with soil had 1.45 (AOR: 1.45 95 % CI: 1.2–3.7) times more likely to have toxoplasmosis infection than those who didn't have contact with soil.

4. Discussions

T. gondii infection is common among humans, though regional variations exist in its prevalence. In general, T. gondii infections are projected to affect one-third of the world's population. In Somalia as a whole, especially the north, there is still an inadequate and absence of information regarding the seropositivity and associated risk factors of Toxoplasma gondii infection among pregnant women, despite the disease's high importance for both animal and public health.

The overall seroprevalence of anti-T. gondii antibodies was 36.4 %

(95 % CI, 31.5–41.2 %). Near findings have been reported in Burkina Faso (31.1 %),²¹ Angola (39.4 %) (23), and Peru (35.8 %).²² This could be due to some similarities in personal hygiene practices, eating habits, immune response status of study participants and so on.

On the other hand, the findings of this study were higher than seroprevalences reported in Ghana (21.5),²³ and Algeria (25 %).²⁴ These differences may result in variations in socioeconomic status characteristics, as well as environmental and geographical factors.

Additionally, prevalence in the present study was lower than other previous studies reported from Mogadishu, Somalia which were 45.2 %¹⁷ and 51.79 %,²⁵ as well as other findings in Cameroon (80 %), and Ethiopia (85.4 %).^{26,27} These variations may result from differences in testing procedures, and difference in sample sizes in each research, and so on.

In this study, most (28.1 %) (108/384 CI95 %: 0.23–0.32) of the seropositive results were IgG *anti-*

T. gondii antibodies, while only 8.3 % (32/384 CI95 %: 0.055–0.11) of the study participants were seropositive for IgM antibodies. These findings are in line with IgG seropositivity (29.9 %) reported in Sri Lanka and in Zambia (0 %)²⁸ or less IgM seropositivity reported in Brazil (5.33 %),²⁹ and Iran (2.7 %).³⁰

In this current study, illiteracy was identified as risk factor for the occurrence of toxoplasmosis among pregnant women. Similar findings have been reported in Brazil.²⁹ On the other hand, the findings of this study have shown significant association between contact with soil and the infection. Similar association have been reported from Eastern China,³¹ Yemen³² and Ethiopia.³³

There have been no significant association between gestational age, history of abortion, age, having domestic cats in a pregnant woman's home, gravidity and toxoplasmosis infection. Similar studies have been reported from Ethiopia, Mexico, Mogadishu Somalia, and Ghana.^{8,9,23,34,35} However, different from these findings have been reported in Morocco, Zambia and China.^{22,23,35} These differences could be due to These distinctions may result from similarities and/or variations in the socioeconomic status, eating habits, personal hygiene practices, testing procedures, environmental and geographical factors, immune response status of the pregnant women, and sample size in each study.

5. Strengths and limitations of the study

This study provides critical baseline data on the prevalence of *Toxoplasma gondii* among pregnant women in the Buhodle district, addressing a significant gap in regional research. The use of competitive ELISA (cELISA) as a diagnostic tool ensured reliable detection of *T. gondii* antibodies, even amidst challenges in accessing advanced diagnostic tools. The study successfully implemented random sampling, allowing for a representative overview of the target population. Additionally, it identified key risk factors, such as education level and soil contact, offering actionable insights that could guide targeted interventions.

However, the study faced several limitations. It was difficult to find local data on toxoplasmosis from the study area, which made contextualizing the findings more challenging. The availability of diagnostic kits and reagents was limited, posing significant logistical hurdles during laboratory work. Blood sample collection was also challenging, as many pregnant women expressed concerns about anemia, leading to some reluctance in providing samples. Furthermore, the cross-sectional design of this study did not allow for establishing causal relationships between *T. gondii* infection and the identified risk factors. Finally, the study did not include molecular characterization of *T. gondii* strains, which could have provided more detailed insights into the parasite's epidemiology.

6. Conclusion and recommendations

In this current study, the overall seroprevalence of 36.4 % (140/384)

among pregnant women age was detected. Twenty-eight-point one percent (28.1 %) anti-*T. gondii* IgG and 8.3 % anti-*T. gondii* IgM seroprevalences, indicators of latent and recent *T. gondii* infections were found. In multivariate analysis, it has also identified that educational status of the participants and contact with soil were important risk factors associated with acquiring *T. gondii* infection among pregnant women at Buhodle healthcare center. The high seroprevalence of toxoplasmosis in the current study recommends the necessity of preventive measures, primarily health education about identified risk factors in order to lower related morbidities and fatalities and lead for improvement of services in antenatal care in hospitals. This high prevalence indicates the presence of disease in the environment.

In bivariate logistic regression analysis identified the following risk factors education, abortion history, presence of cats at home, soil exposure, became candidates for multivariable logistic regression ($p \leq 0.25$). However, only illiterate and soil have shown association in multivariate regression. The odds of pregnant women who were illiterate 4.036 more likely to develop toxoplasmosis. Additionally, the odds of pregnant had contact with soil were 1.45 more likely to develop toxoplasmosis.

Based on the above conclusion, the following recommendations were forwarded.

- Educational and awareness creation towards pregnant mother about the zoonotic potential of the disease
- Checkups should be performed at the antenatal care clinic with the aim to prevent maternal morbidity and congenital toxoplasmosis.
- Further studies should be done in the future to determine the important risk factors using cost effective diagnostic tests to investigate toxoplasmosis
- The concerned bodies, particularly the health bureau have to prioritize the importance of zoonosis in the area, and apply the appropriate actions towards preventive measures.
- Avoiding continuous exposure to soil and/or practicing proper cat waste management in the area

Authors' contributions

Mohamed Osman is substantial contribution from inception of idea, proposal development, data collection, analysis, interpretation and final write up and also contributed writing the manuscript. Abdilahi Adan contributed to overall work designation, statistical analysis of the data and helped with the interpretation of results. Abdilahi Abdiweli Md. Al-Mamun and Marufa Alam, developed, edited the manuscript and finally revised. All authors read and approved the final version of the manuscript.

Ethical consideration

The study protocol was reviewed and approved by the Research Ethics and Review Committee of Jigjiga University College of Health and Medical Sciences with Ref. No IHRR03/2024. Official permission was also obtained from the respective body at Buhodle General Hospital and other healthcare facilities by providing letters from the University. Verbal and written informed consent were obtained from each participant after informing about the purpose of the study, the procedure, the risk, benefit and also their right as study participants. The blood sample was collected by strictly following the standard operational procedure for blood sample collection and by minimizing any discomfort. Any information was kept confidentially and anonymous test was done. For those participants whose laboratory result was positive and clinically significant, the investigator will consult a physician and communicate for further management.

Human consent

Written informed consent was obtained from the patient for publication and any accompanying images. A copy of the written consent is available for review by the Editor-in-Chief of this journal on request.

Availability of data and materials

The datasets generated and/or analyzed during the current study are available from the corresponding author upon reasonable request. Specific data and materials will be provided in accordance with the ethical guidelines and institutional policies.

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Declaration of interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: No conflict of interest to declare.

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